

## Role-Based Access Control

# Lab 01: Role-Based Access Control

## Student lab manual

### Lab scenario

You have been asked to create a proof of concept showing how Azure users and groups are created. Also, how role-based access control is used to assign roles to groups. Specifically, you need to:

- Create a Senior Admins group containing the user account of Joseph Price as its member.
- Create a Junior Admins group containing the user account of Isabel Garcia as its member.
- Create a Service Desk group containing the user account of Dylan Williams as its member.
- Assign the Virtual Machine Contributor role to the Service Desk group.

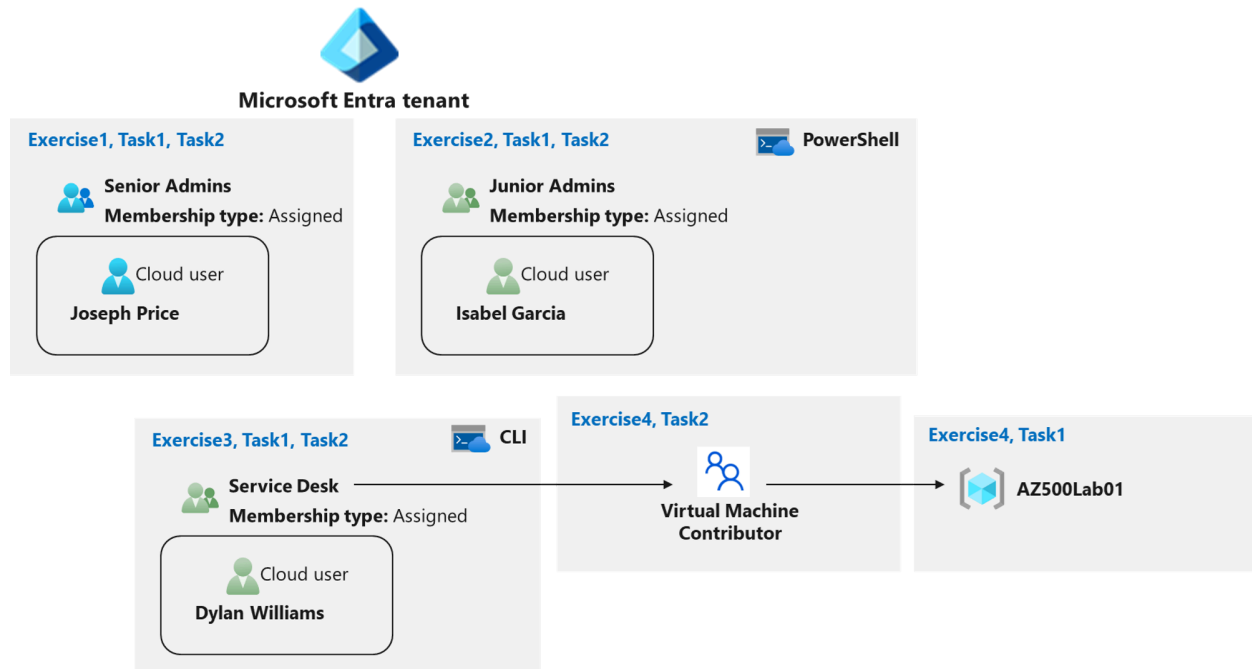
For all the resources in this lab, we are using the **East US** region. Verify with your instructor this is the region to use for class.

### Lab objectives

In this lab, you will complete the following exercises:

- Exercise 1: Create the Senior Admins group with the user account Joseph Price as its member (the Azure portal).
- Exercise 2: Create the Junior Admins group with the user account Isabel Garcia as its member (PowerShell).
- Exercise 3: Create the Service Desk group with the user Dylan Williams as its member (Azure CLI).
- Exercise 4: Assign the Virtual Machine Contributor role to the Service Desk group.

## Role-Based Access Control architecture diagram



## Instructions

Exercise 1: Create the Senior Admins group with the user account Joseph Price as its member.

Estimated timing: 10 minutes

In this exercise, you will complete the following tasks:

- Task 1: Use the Azure portal to create a user account for Joseph Price.
- Task 2: Use the Azure portal to create a Senior Admins group and add the user account of Joseph Price to the group.

### Task 1: Use the Azure portal to create a user account for Joseph Price

In this task, you will create a user account for Joseph Price.

1. Start a browser session and sign-in to the Azure portal  
<https://portal.azure.com/>.  
Sign in to the Azure portal using an account that has the Owner or Contributor role

in the Azure subscription you are using for this lab and the Global Administrator role in the Microsoft Entra tenant associated with that subscription.

2. In the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Microsoft Entra ID** and press the **Enter** key.
3. On the **Overview** blade of the Microsoft Entra ID tenant, in the **Manage** section, select **Users**, and then select **+ New user**.  
**Steps 3-8** are for reference only, as this user has already been created for you. If you wish, you may review them but the **New User** button will be greyed out. This is expected in this **Cloudslice** lab and you may proceed to **Task 2**.
4. On the **New User** blade, ensure that the **Create user** option is selected, and specify the following settings:

| Setting   | Value        |
|-----------|--------------|
| User name | Joseph       |
| Name      | Joseph Price |

5. Click on the copy icon next to the **User name** to copy the full user.
6. Ensure that the **Auto-generate** password is selected, select the **Show password** checkbox to identify the automatically generated password. You would need to provide this password, along with the user name to Joseph.
7. Click **Create**.
8. Refresh the **Users | All users** blade to verify the new user was created in your Microsoft Entra tenant.

**Task2: Use the Azure portal to create a Senior Admins group and add the user account of Joseph Price to the group.**

In this task, you will create the *Senior Admins* group, add the user account of Joseph Price to the group, and configure it as the group owner.

1. In the Azure portal, navigate back to the blade displaying your Microsoft Entra ID tenant.
2. In the **Manage** section, click **Groups**, and then select **+ New group**.
3. On the **New Group** blade, specify the following settings (leave others with their default values):

| Setting         | Value                 |
|-----------------|-----------------------|
| Group type      | Security              |
| Group name      | Senior Admins63227258 |
| Membership type | Assigned              |

4. Click the **No owners selected** link, on the **Add owners** blade, select **Joseph Price**, Joseph-63227258@LODSPRODMSLEARNMCA.onmicrosoft.com and click **Select**.
5. Click the **No members selected** link, on the **Add members** blade, select **Joseph Price**, Joseph-63227258@LODSPRODMSLEARNMCA.onmicrosoft.com and click **Select**.
6. Back on the **New Group** blade, click **Create**.  
Result: You used the Azure Portal to create a user and a group, and assigned the user to the group.

## Exercise 2: Create a Junior Admins group containing the user account of Isabel Garcia as its member.

Estimated timing: 10 minutes

In this exercise, you will complete the following tasks:

- Task 1: Use PowerShell to create a user account for Isabel Garcia.
- Task 2: Use PowerShell to create the Junior Admins group and add the user account of Isabel Garcia to the group.

### Task 1: Use PowerShell to create a user account for Isabel Garcia.

In this task, you will create a user account for Isabel Garcia by using PowerShell.

1. **Open the Cloud Shell** by clicking the **Cloud Shell icon** in the top-right corner of the Azure portal.
2. If prompted, select **No storage account required**, select the name of your subscription, and then select **Apply**. This is required *only the first time* you launch the Cloud Shell.
3. In the Cloud Shell pane, **ensure PowerShell is selected** from the drop-down menu in the upper-left corner. (Note: In the new Cloud Shell, this will say: **-->Switch to Bash**)  
To paste copied text into the Cloud Shell, right-click within the pane window and select **Paste**. Alternatively, you can use the **Shift+Insert** key combination.
4. In the PowerShell session within the Cloud Shell pane, run the following to create a password profile object:
5. 

```
$passwordProfile = New-Object -TypeName Microsoft.Open.AzureAD.Model.PasswordProfile
```
6. In the PowerShell session within the Cloud Shell pane, run the following to set the value of the password within the profile object:
7. 

```
$passwordProfile.Password = "Pa55w.rd1234"
```

8. In the PowerShell session within the Cloud Shell pane, run the following to connect to Microsoft Entra ID:
9. `Connect-AzureAD`
10. In the PowerShell session within the Cloud Shell pane, run the following to identify the name of your Microsoft Entra tenant:
11. `$domainName = ((Get-AzureAdTenantDetail).VerifiedDomains)[0].Name`
12. In the PowerShell session within the Cloud Shell pane, run the following to create a user account for Isabel Garcia:  
**Step 8** is for reference only, and can be skipped as this user has already been created for you. If you wish, you may run the command but you will receive the error **New-AzureADUser: Error occurred while executing NewUser**. This is expected in this **Cloudslice** lab and you may proceed to the next **Step**.
13. `New-AzureADUser -DisplayName 'Isabel Garcia' -PasswordProfile $passwordProfile -UserPrincipalName "Isabel@$domainName" -AccountEnabled $true -MailNickName 'Isabel'`
14. In the PowerShell session within the Cloud Shell pane, run the following to list Microsoft Entra ID users (the accounts of Joseph and Isabel should appear on the listed):
15. `Get-AzureADUser -All $true | Where-Object {$_.UserPrincipalName -like "Isabel-63227258*"}`

**Task2: Use PowerShell to create the Junior Admins group and add the user account of Isabel Garcia to the group.**

In this task, you will create the Junior Admins group and add the user account of Isabel Garcia to the group by using PowerShell.

1. In the same PowerShell session within the Cloud Shell pane, run the following to **create a new security group** named Junior Admins:
2. `New-AzureADGroup -DisplayName 'Junior Admins6322725863227258' -MailEnabled $false -SecurityEnabled $true -MailNickName JuniorAdmins`
3. In the PowerShell session within the Cloud Shell pane, run the following to **list groups** in your Microsoft Entra tenant (the list should include the Senior Admins63227258 and Junior Admins63227258 groups)
4. `Get-AzureADGroup`
5. In the PowerShell session within the Cloud Shell pane, run the following to **obtain a reference** to the user account of Isabel Garcia:
6. `$user = Get-AzureADUser -Filter "UserPrincipalName eq 'Isabel-63227258@LODSPRODMSLEARNMCA.onmicrosoft.com'"`
7. In the PowerShell session within the Cloud Shell pane, run the following to add the user account of Isabel to the Junior Admins6322725863227258 group:

8. `Add-AzADGroupMember -MemberUserPrincipalName $user.userPrincipalName -TargetGroupDisplayName "Junior Admins6322725863227258"`
9. In the PowerShell session within the Cloud Shell pane, run the following to verify that the Junior Admins6322725863227258 group contains the user account of Isabel:
10. `Get-AzADGroupMember -GroupDisplayName "Junior Admins6322725863227258"`
11. Result: You used PowerShell to create a user and a group account, and added the user account to the group account.

## Exercise 3: Create a Service Desk group containing the user account of Dylan Williams as its member.

Estimated timing: 10 minutes

In this exercise, you will complete the following tasks:

- Task 1: Use Azure CLI to create a user account for Dylan Williams.
- Task 2: Use Azure CLI to create the Service Desk group and add the user account of Dylan to the group.

### Task 1: Use Azure CLI to create a user account for Dylan Williams.

In this task, you will create a user account for Dylan Williams.

1. In the drop-down menu in the upper-left corner of the Cloud Shell pane, select **Bash** (or: **Switch to Bash**), and, when prompted, click **Confirm**.
2. In the Bash session within the Cloud Shell pane, run the following to identify the name of your Microsoft Entra tenant:
3. `DOMAINNAME=$(az ad signed-in-user show --query 'userPrincipalName' | cut -d '@' -f 2 | sed 's/\\/\\/')`
4. In the Bash session within the Cloud Shell pane, run the following to create a user, Dylan Williams. Use *yourdomain*.  
**Step 3** is for reference only, as this user has already been created for you. If you wish, you may run the command but you will receive the error **Insufficient privileges to complete the operation**. This is expected in this **Cloudslice** lab and you may proceed to the next **Step**.
5. `az ad user create --display-name "Dylan Williams" --password "Pa55w.rd1234" --user-principal-name Dylan@$DOMAINNAME`

6. In the Bash session within the Cloud Shell pane, run the following to list Microsoft Entra ID user accounts (the list should include user accounts of Joseph, Isabel, and Dylan)
7. 

```
az ad user list --output table | grep 63227258
```

## Task 2: Use Azure CLI to create the Service Desk group and add the user account of Dylan to the group.

In this task, you will create the Service Desk group and assign Dylan to the group.

1. In the same Bash session within the Cloud Shell pane, run the following to create a new security group named Service Desk.
2. 

```
az ad group create --display-name "Service Desk63227258" --mail-nickname "ServiceDesk"
```
3. In the Bash session within the Cloud Shell pane, run the following to list the Microsoft Entra ID groups (the list should include Service Desk63227258, Senior Admins63227258, and Junior Admins63227258 groups):
4. 

```
az ad group list -o table
```
5. In the Bash session within the Cloud Shell pane, run the following to obtain a reference to the user account of Dylan Williams:
6. 

```
USER=$(az ad user list --filter "UserPrincipalName eq 'Dylan-63227258@LODSPRODMSLEARNMCA.onmicrosoft.com'")
```
7. In the Bash session within the Cloud Shell pane, run the following to obtain the objectId property of the user account of Dylan Williams:
8. 

```
OBJECTID=$(echo $USER | jq '.[].id' | tr -d '"')
```
9. In the Bash session within the Cloud Shell pane, run the following to add the user account of Dylan to the Service Desk63227258 group:
10. 

```
az ad group member add --group "Service Desk63227258" --member-id $OBJECTID
```
11. In the Bash session within the Cloud Shell pane, run the following to list members of the Service Desk63227258 group and verify that it includes the user account of Dylan:
12. 

```
az ad group member list --group "Service Desk63227258"
```
13. Close the Cloud Shell pane.  
Result: Using Azure CLI you created a user and a group accounts, and added the user account to the group.

## Exercise 4: Assign the Virtual Machine Contributor role to the Service Desk group.

Estimated timing: 10 minutes

In this exercise, you will complete the following tasks:

- Task 1: Create a resource group.
- Task 2: Assign the Service Desk Virtual Machine Contributor permissions to the resource group.

## Task 1: Create a resource group

1. In the Azure portal, in the **Search resources, services, and docs** text box at the top of the Azure portal page, type **Resource groups** and press the **Enter** key.
2. On the **Resource groups** blade, click **+ Create** and specify the following settings:

| Setting             | Value                               |
|---------------------|-------------------------------------|
| Subscription name   | the name of your Azure subscription |
| Resource group name | AZ500Lab01                          |
| Location            | East US                             |

3. Click **Review + create** and then **Create**.  
Wait for the resource group to deploy. Use the **Notification** icon (top right) to track progress of the deployment status.
4. Back on the **Resource groups** blade, refresh the page and verify your new resource group appears in the list of resource groups.

## Task 2: Assign the Service Desk Virtual Machine Contributor permissions.

1. On the **Resource groups** blade, click the **AZ500LAB01** resource group entry.
2. On the **AZ500Lab01** blade, click **Access control (IAM)** in the middle pane.
3. On the **AZ500Lab01 | Access control (IAM)** blade, click **+ Add** and then, in the drop-down menu, click **Add role assignment**.
4. On the **Add role assignment** blade, complete each of the following settings before clicking **Next**:

After completing all the steps, click **Next**.

| Setting                               | Value                             |
|---------------------------------------|-----------------------------------|
| Role in the search tab                | Virtual Machine Contributor       |
| Assign access to (Under Members Pane) | User, group, or service principal |
| Select (+Select Members)              | Service Desk63227258              |

5. Click **Review + assign** twice to create the role assignment.
6. From the **Access control (IAM) blade**, select **Check access**.
7. On the **AZ500Lab01 | Access control (IAM)** blade, on the **Check access** tab, in the **Search by name or email address** text box, check access for Dylan-63227258@LODSPRODMSLEARNMCA.onmicrosoft.com.
8. In the list of search results, select the user account of Dylan Williams and, on the **Dylan Williams assignments - AZ500Lab01** blade, view the newly created assignment.  
NOTE - Always remember to check both the **Active** and **Eligible** tabs when confirming RBAC assignments.
9. Close the **Dylan Williams assignments - AZ500Lab01** blade.
10. Repeat the same last two steps to check access for Joseph-63227258@LODSPRODMSLEARNMCA.onmicrosoft.com.  
Result: You have assigned and checked RBAC permissions.

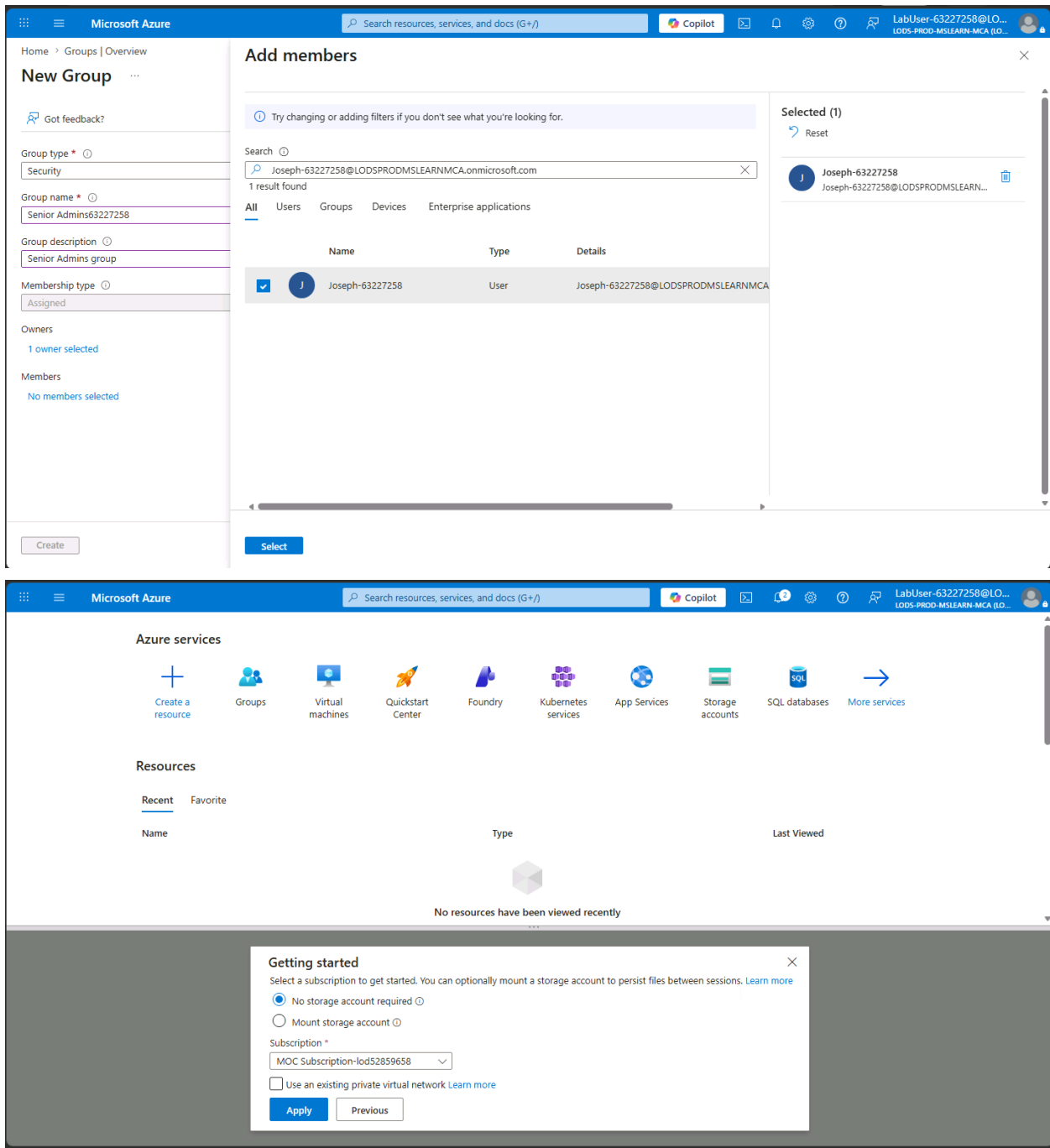
## Clean up resources

Remember to remove any newly created Azure resources that you no longer use. Removing unused resources ensures you will not incur unexpected costs.

1. In the Azure portal, open the Cloud Shell by clicking the first icon in the top right of the Azure Portal.
2. In the drop-down menu in the upper-left corner of the Cloud Shell pane, select **PowerShell**, and, when prompted, click **Confirm**.
3. In the PowerShell session within the Cloud Shell pane, run the following to remove the resource group you created in this lab:
4. 

```
Remove-AzResourceGroup -Name "AZ500LAB01" -Force -AsJob
```
5. Close the **Cloud Shell** pane.

Screenshots



```
Microsoft Azure
Search resources, services, and docs (G+)
Copilot
LabUser-63227258@LO...
LODS-PROD-MSLEARN-MCA (LO...

Switch to Bash Restart Manage files New session Editor Open in VS Code for the Web Web preview Settings Help

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.

MOTD: Share your feedback and help us improve Cloud Shell: https://aka.ms/cloudshell/feedback

VERBOSE: Authenticating to Azure ...
PS /home/labuser-63227258> $passwordProfile = New-Object -TypeName Microsoft.Open.AzureAD.Model.PasswordProfile
PS /home/labuser-63227258> $passwordProfile.Password = "Pa55w.rd1234"
PS /home/labuser-63227258> Connect-AzureAD
PS /home/labuser-63227258> $domainName = ((Get-AzureAdTenantDetail).VerifiedDomains)[0].Name
PS /home/labuser-63227258> New-AzureADUser -DisplayName 'Hari Madaraju' -PasswordProfile $passwordProfile -UserPrincipalName "Hari@$domainName" -AccountEnabled $true -MailNickName 'Hari'
New-AzureADUser: Error occurred while executing NewUser
Code: Authorization_RequestDenied
Message: Insufficient privileges to complete the operation.
RequestId: e34b72c3-e5ef-4c78-9de4-e1a80df12f
DateTimeStamp: Sun, 05 Jul 2026 09:57:10 GMT
HttpStatusCode: Forbidden
HttpStatusDescription: Forbidden
HttpResponseStatus: Completed
PS /home/labuser-63227258> New-AzureADUser -DisplayName 'Isabel Garcia' -PasswordProfile $passwordProfile -UserPrincipalName "Isabel@$domainName" -AccountEnabled $true -MailNickName 'Isabel'
New-AzureADUser: Error occurred while executing NewUser
Code: Authorization_RequestDenied
Message: Insufficient privileges to complete the operation.
RequestId: cf67bd4f-c875-4cc4-83ff-c31596a8ca64
DateTimeStamp: Sun, 05 Jul 2026 09:57:26 GMT
HttpStatusCode: Forbidden
HttpStatusDescription: Forbidden
HttpResponseStatus: Completed
PS /home/labuser-63227258> Get-AzureADUser -All $true | Where-Object {$_.UserPrincipalName -like "Isabel-63227258*"}

ObjectId      DisplayName      UserPrincipalName      UserType
-----
1f61379c-bfa0-4b11-9512-d875e2f9db55 Isabel-63227258 Isabel-63227258@LODSPRODMSLEARNMCA.onmicrosoft.com Member

PS /home/labuser-63227258>
```

```
Microsoft Azure
Search resources, services, and docs (G+)
Copilot
LabUser-63227258@LO...
LODS-PROD-MSLEARN-MCA (LO...

Switch to Bash Restart Manage files New session Editor Open in VS Code for the Web Web preview Settings Help

PS /home/labuser-63227258> New-AzureADGroup -DisplayName 'Junior Admins6322725863227258' -MailEnabled $false -SecurityEnabled $true -MailNickName JuniorAdmins

ObjectId      DisplayName      Description
-----
c81d3ea3-fd5f-4fac-9bef-b64752c98720 Junior Admins6322725863227258

PS /home/labuser-63227258> Get-AzureADGroup

ObjectId      DisplayName      Description
-----
03346c7e-2c6d-4d4c-a26d-1debfd4196e2 62281173-AppLogExaminers
039cfc5b-061c-467c-bcb2-01fa105203da Senior Admins61052486
05dcbfcd-17df-49b1-a791-98873862fe35 student
0a81c104-f079-4dd0-8775-8a32c76e3d3f TechMasters Fabric Labs
0d3cb1d0-1287-455e-a2c7-8a4c936bd7ef Senior Admins60943236
0e67ba6e-06f7-4cc6-b408-e24b2ec62111 Service Desk40153487
0f4dc777-8d34-455f-97bd-29d6522d6703 SeniorAdmin Group
0fb4022c-97e0-412f-92a5-ab610eb02ff3 Senior Admins61113618
110e3519-2570-4ef3-8d8f-7fa4ee7fb226 Junior Admins36087824
1168a676-15f4-490e-8d84-fa906f6e8931 SOC Analysts
118c7c5e-1b88-4255-b46d-252e66352c8 Senior Admins60869711
11e33e32-98d6-4db9-b5b4-c9d9bc60408d Senior Admins61138624
1245490f-b476-4112-8028-839aea8ad6 EhabT
12840017-dd46-412c-a079-1006ac808c98 ConexionPrueba
12bd7a6c-1f5d-4161-a409-62c3c454a279 Service Desk61056902
1679b5e3-d68d-4bcd-8c9e-115f02f236be Service Desk60943236
1745eb3d-8822-4ee4-a342-cd5e310ae2c3 Service Desk61641593
19cccbec-90f2-4de3-8f9b-95f835899147 Senior Admins38936808
1aad08af-5a79-464a-badd-d38defcc06 Service Desk61394708
1db3491e-744c-48e4-abad-49a344c1799b Senior Admins60277400
1f697dc2-7532-43dc-a863-015f2bc2f9a9 62253258-AppLogExaminers
1f6e2e93-619d-4f38-82ea-59ebdd30757e 60380453-AppLogExaminers
202d8319-9b3e-42d1-9bbd-0df337016b3d Senior Admins63227258
20ad077e-861f-46c0-9b91-40e445039f59 SkillableLabContentAdmins
20f8080a-630d-4e76-9725-c424b8f93a2d Senior Admins61641595
216083d1-697d-49b2-a171-dc066a23bd90 Junior Admins602725860272528
```

```
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
LabUser-63227258@LO...
LODS-PROD-MSLEARN-MCA (LO...)

Switch to Bash Restart Manage files New session Editor Open in VS Code for the Web Web preview Settings Help

PS /home/labuser-63227258> Add-AzADGroupMember -MemberUserPrincipalName $user.userPrincipalName -TargetGroupDisplayName "Junior Admins6322725863227258"
WARNING: This cmdlet is using a preview API version and is subject to breaking change in a future release.
PS /home/labuser-63227258> Get-AzADGroupMember -GroupDisplayName "Junior Admins6322725863227258"
WARNING: This cmdlet is using a preview API version and is subject to breaking change in a future release.

AccountEnabled : True
AgeGroup       :
ApproximateLastSignInDateTime :
BusinessPhone  : {}
City           :
CompanyName    :
ComplianceExpirationDateTime :
ConsentProvidedForMinor      :
Country        :
CreatedDateTime : 7/5/2026 9:25:46 AM
CreationType    :
DeletedDateTime :
Department     :
DeviceVersion   :
DisplayName     : Isabel-63227258
EmployeeHireDate :
EmployeeId      :
EmployeeOrgData : {}
EmployeeType    :
ExternalUserState :
ExternalUserStateChangeDateTime :
FaxNumber       :
GivenName       :
Id              : 1f61379c-bfa0-4b11-9512-d875e2f9db55
Identity        : {{
  "issuer": "LODSPRODMSLEARNMCA.onmicrosoft.com",
  "issuerAssignedId": "Isabel-63227258@LODSPRODMSLEARNMCA.onmicrosoft.com",
  "signInType": "userPrincipalName"
}}
ImAddress       : {}
```

```
Microsoft Azure
Search resources, services, and docs (G+/)
Copilot
LabUser-63227258@LO...
LODS-PROD-MSLEARN-MCA (LO...)

Switch to PowerShell Restart Manage files New session Editor Open in VS Code for the Web Web preview Settings Help

Requesting a Cloud Shell.Succeeded.
Connecting terminal...

Subscription used to launch your CloudShell f15a207c-1cda-4a72-a0d0-a216bf5a4463 is not registered to Microsoft.CloudShell Namespace. Please follow these instructions "https://aka.ms/RegisterCloudShell" to register. In future, unregistered subscriptions will have restricted access to CloudShell service.

Your Cloud Shell session will be ephemeral so no files or system changes will persist beyond your current session.
labuser-63227258 [ ~ ]$ DOMAINNAME=$(az ad signed-in-user show --query 'userPrincipalName' | cut -d '@' -f 2 | sed 's/\//')
labuser-63227258 [ ~ ]$ az ad user create --display-name "Dylan Williams" --password "Pa$5w.rd1234" --user-principal-name Dylan@$DOMAINNAME
Insufficient privileges to complete the operation.
labuser-63227258 [ ~ ]$ az ad user list --output table | grep 63227258
Dylan-63227258                                Dylan-63227258@LODSPRODMSLEARNMCA.onmicrosoft.com
Isabel-63227258                                Isabel-63227258@LODSPRODMSLEARNMCA.onmicrosoft.com
Joseph-63227258                               Joseph-63227258@LODSPRODMSLEARNMCA.onmicrosoft.com
LabUser-63227258                              LabUser-63227258@LODSPRODMSLEARNMCA.onmicrosoft.com
labuser-63227258 [ ~ ]$ az ad group create --display-name "Service Desk63227258" --mail-nickname "ServiceDesk"
{
  "@odata.context": "https://graph.microsoft.com/v1.0/$metadata#groups/$entity",
  "classification": null,
  "createdDateTime": "2026-07-05T10:02:20Z",
  "creationOptions": [],
  "deletedDateTime": null,
  "description": null,
  "displayName": "Service Desk63227258",
  "expirationDateTime": null,
  "groupTypes": [],
  "id": "ffdbc009-8198-45b0-98f3-1f1db99a2a1b",
  "infoCatalogs": [],
  "isAssignableToRole": null,
  "mail": null,
  "mailEnabled": false,
  "mailNickname": "ServiceDesk",
  "membershipRule": null,
  "membershipRuleProcessingState": null,
  "onPremisesDomainName": null,
  "onPremisesLastSyncDateTime": null,
  "onPremisesNetBiosName": null,
  "onPremisesProvisioningErrors": [],

```

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

LabUser-63227258@L.O...  
LOOS-PROD-S0LEARN-MCA ID

Azure services

Switch to PowerShell

Restart

Manage files

New session

Editor

Open in VS Code for the Web

Web preview

Settings

Help

```
}
labuser-63227258 [ ~ ]$ az ad group list -o table
CreatedDateTime      DisplayName      SecurityIdentifier      Description      MailEnabled      MailNickname      RenewedDate
-----
2026-06-01T03:54:52Z 62281173-AppLogExaminers  S-1-12-1-53767294-1296837741-3944574370-3801498109  False      62281173-AppLogExaminers  2026-06-01
T03:54:52Z      True      Senior Admins61052486  S-1-12-1-60619867-1182533148-4194415292-3657650704  False      8f83a485-d      2026-04-20
T13:33:47Z      True      student  S-1-12-1-98353101-1236342751-2274922919-905863736  student demo  False      a2ba8d72-6      2024-06-24
T17:27:51Z      True      TechMasters Fabric Labs  S-1-12-1-176275716-1305538681-847934855-1060990663  True      Private  2025-02-26
T19:02:58Z      True      Senior Admins60943236  S-1-12-1-222881488-1163793031-1284163490-4023872403  False      286482ca-2      2026-04-16
T11:48:56Z      True      Service Desk40153487  S-1-12-1-241678958-1288046327-1273104564-287426094  False      ServiceDesk  2024-04-28
T23:02:18Z      True      SeniorAdmin Group  S-1-12-1-25675575-1163909181-3593059735-57097554  Senior group account  False      cf9e2a03-7      2024-03-04
T16:11:34Z      True      Senior Admins61113619  S-1-12-1-263455276-1093638112-1638638994-4079988750  False      6c605708-e      2026-04-21
T20:47:15Z      True      Junior Admins36087824  S-1-12-1-286143769-1324557680-3833569165-649232366  False      JuniorAdminsGet-AzureADGroup  2023-12-04
T19:45:54Z      True      SOC Analysts  S-1-12-1-294419550-1112873864-774204852-3360842598  Security Operations Center Analysts  False      15fdb4ab-c      2026-03-25
T07:22:09Z      True      Senior Admins60869711  S-1-12-1-300105266-1304008918-3653874869-2369086524  False      6002c5a6-8      2026-04-14
T15:52:31Z      True      Senior Admins61138624  S-1-12-1-4245402289-1255446304-1695643315-3428679634  False      d413823d-c      2026-04-22
T13:14:44Z      True      EhabT  S-1-12-1-4252966397-1280635835-2599326143-1621888481  test inh  False      0ef99f19-8      2023-11-22
T11:30:21Z      True      ConexlonPrueba  S-1-12-1-4292591625-1169195416-488633240-455776953  True      ConexlonPrueba  2025-04-10
T05-04-10T07:19:58Z
```

Switch to PowerShell

Restart

Manage files

New session

Editor

Open in VS Code for the Web

Web preview

Settings

Help

```
T19:09:27Z      True      Used to provision SCIM-based access to shared Databricks Workspace  S-1-12-1-4141867104-1118791682-3370017941-1361057727  False      ServiceDesk  2026-05-09
2026-05-09T11:31:18Z      Service Desk61641595  S-1-12-1-4168769198-1196186328-2537832339-1563293086  False      JuniorAdmins  2026-04-16
T11:31:18Z      True      Junior Admins6094323660943236  S-1-12-1-4172792103-1119149146-777835656-3989265685  False      62263603-AppLogExaminers  2026-05-31
2026-04-16T11:54:01Z      True      groupix  S-1-12-1-4227233159-1172539672-2453748108-3479602657  False      e27e9d5a-3      2024-10-10
T07:53:26Z      True      _Skillable - Microsoft 365 Copilot  S-1-12-1-4238377029-1291666600-1007463063-3022569438  False      3db2795c-4      2025-09-11
2024-10-10T20:35:10Z      True      Senior Admins60434917  S-1-12-1-4252966397-1280635835-2599326143-1621888481  False      2e5d8371-1      2026-03-30
T20:35:10Z      True      Service Desk63227258  S-1-12-1-4292591625-1169195416-488633240-455776953  False      ServiceDesk  2026-07-05
T10:03:15Z      True      T10:02:20Z      True
labuser-63227258 [ ~ ]$ USER=$(az ad user list --filter "UserPrincipalName eq 'Dylan-63227258@LOOSPRODMSLEARNMCA.onmicrosoft.com'")
labuser-63227258 [ ~ ]$ OBJECTID=$(echo $USER | jq '.[].id' | tr -d '"')
labuser-63227258 [ ~ ]$ az ad group member add --group "Service Desk63227258" --member-id $OBJECTID
labuser-63227258 [ ~ ]$ az ad group member list --group "Service Desk63227258"

[
  {
    "@odata.type": "#microsoft.graph.user",
    "businessPhones": [],
    "displayName": "Dylan-63227258",
    "givenName": null,
    "id": "18b81146-983c-489a-ac97-9f9761f3dbc0",
    "jobTitle": null,
    "mail": null,
    "mobilePhone": null,
    "officeLocation": null,
    "preferredLanguage": null,
    "surname": null,
    "userPrincipalName": "Dylan-63227258@LOOSPRODMSLEARNMCA.onmicrosoft.com"
  }
]
labuser-63227258 [ ~ ]$
```

Microsoft Azure

Search resources, services, and docs (G+)

Copilot

LabUser-63227258@LO...  
LODS-PROD-MSLEARN-MCA (LO...

Home

Create a resource group

BasicsTagsReview + create

**Resource group** - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription \*

Resource group name \*

Region \*

Previous

Next

Review + create

Home > AZ500Lab01

AZ500Lab01 | Access control (IAM)

OverviewActivity logAccess control (IAM)TagsResource visualizerEventsSettingsCost ManagementMonitoringAutomationHelp

Search

Add

Download role assignments

Edit columns

Refresh

Delete

Feedback

Add role assignment

Add custom role

Assignments

Roles

Deny assignments

Looking for the previous check access view? [Click here.](#)

Check access

Review the level of access a user, group, service principal, or managed identity has to this resource. [Learn more](#)

LabUser-63227258@LODSPRODMSLEARNMCA.onmicrosoft.com Assignments

| Active | Eligible | Deny |
|--------|----------|------|
| 1      | 0        | 0    |

Search by role name or membership

| Role Name                          | Scope                    | Membership       | Condition | Action |
|------------------------------------|--------------------------|------------------|-----------|--------|
| Active permanent assignments (1)   |                          |                  |           |        |
| LOD Owner                          | Subscription (Inherited) | LabUser-63227258 | Yes       | -      |
| Active time-bound assignments (0)  |                          |                  |           |        |
| Eligible permanent assignments (0) |                          |                  |           |        |

Add or remove favorites by pressing Ctrl+Shift+F

## Add role assignment

Role Members Conditions Assignment type Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Copilot can help pick a role

Job function roles Privileged administrator roles

Grant access to Azure resources based on job function, such as the ability to create virtual machines.

Virtual Machine Contributor

Type: All Category: All

| Name                                           | Description                                                                                                                                               | Type        | Category |
|------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|----------|
| Classic Virtual Machine Contributor            | Lets you manage classic virtual machines, but not access to them, and not the virtual network or storage account they're connected to.                    | BuiltInRole | Compute  |
| Desktop Virtualization Power On Contributor    | Provide permission to the Azure Virtual Desktop Resource Provider to start virtual machines.                                                              | BuiltInRole | None     |
| Desktop Virtualization Power On Off Contrib... | Provide permission to the Azure Virtual Desktop Resource Provider to start and stop virtual machines.                                                     | BuiltInRole | None     |
| Desktop Virtualization Virtual Machine Cont... | This role is in preview and subject to change. Provide permission to the Azure Virtual Desktop Resource Provider to create, delete, update, start, a...   | BuiltInRole | None     |
| Service Fabric Cluster Contributor             | Manage your Service Fabric Cluster resources. Includes clusters, application types, application type versions, applications, and services. You will ne... | BuiltInRole | None     |
| Virtual Machine Contributor                    | Lets you manage virtual machines, but not access to them, and not the virtual network or storage account they're connected to.                            | BuiltInRole | Compute  |

Showing 1 - 6 of 6 results.

Review + assign Previous Next

Feedback

## Add role assignment

Role Members Conditions Assignment type Review + assign

Showing a filtered list of roles because your permissions include a condition. [Learn more](#)  
[View my access](#)

Selected role Virtual Machine Contributor

Assign access to ☒ User, group, or service principal  
☐ Managed identity

Members + Select members

| Name                | Object ID | Type |
|---------------------|-----------|------|
| No members selected |           |      |

Description Optional

Review + assign Previous Next

## Select members

Service Desk63227258

SD Service Desk63227258  
ffabc009-8198-45b0-98f3-1f1db99a2a1b

Selected members:

SD Service Desk63227258  
ffabc009-8198-45b0-98f3-1f1db99a2a1b

Select Close

## Add role assignment



Role   Members   Conditions   **Assignment type**   Review + assign

If you have Microsoft Entra Privileged Identity Management (PIM), you can use eligible assignments to provide just-in-time access to role. Users with eligible and/or time-bound assignments must have a valid license. [Learn more](#)

**Selected role**

Virtual Machine Contributor

**Assignment type**

- ☒ **Eligible (Recommended)**  
Member must activate to use this role for a limited period of time.
- ☐ **Active**  
Member can use this role at any time.

**Assignment duration**

- ☐ **Permanent**  
Assignment has no end date or time.
- ☒ **Time bound**  
Assignment has an end date and time.

**Start date and time\***

07/05/2026 3:09 AM

**End date and time\*** ⓘ

07/05/2027 3:09 AM

▼ **Configure Privileged Identity Management (PIM) policy**

**Review + assign**

Previous

Next

Feedback

### AZ500Lab01 | Access control (IAM)

Search   ◯   ◀   + Add   ▾   Download role assignments

Overview

Activity log

**Access control (IAM)**

Tags

Resource visualizer

Events

Settings

Cost Management

Monitoring

Automation

Help

Check access   Role assignments   Roles

Looking for the previous check access view? Click here

**Check access**

Review the level of access a user, group, service principal, or application has to a resource.

**Check access**

LabUser-63227258@LODSF

Active ⓘ   Eligible ⓘ   Denied ⓘ

Search by role name or membership

Role Name ↑↓   Scope ↑↓

▼ Active permanent assignments (1)

LOD Owner   Subscriptions

Active time-bound assignments (0)

Eligible permanent assignments (0)

### Dylan-63227258 assignments - AZ500Lab01

Current role assignments   **Eligible assignments**

Activate a role to perform operations that require elevated permissions. The role will remain active for a limited time.

Activate role

Search by role name

| Role                        | Scope         | Membership | End time          |
|-----------------------------|---------------|------------|-------------------|
| Virtual Machine Contributor | This resource | Group      | 7/5/2027, 3:09 AM |